SCR Hub board (IGVC 2027)

The power, CAN and E-Stop distribution board for Sooner Competitive Robotics' IGVC 2027 robot, with current sensing on every port, designed in Altium.

Features

- 12 V input, 5–8 A total, distributed to seven 6-pin Molex board ports
- Power, CAN bus and the hardware E-Stop line on every port
- INA220 current and voltage sensing on each port
- P-MOSFET reverse-polarity protection (DMP4015SK3Q-13)
- AP63301 synchronous buck for the 3.3 V rail, rated to 32 V input
- Proposed STM32C092 microcontroller with a MAX3051 CAN transceiver
- 8 MHz Abracon ABLS crystal, ±30 ppm, for CAN timing
- SN74AHC244 buffer gives each port its own E-Stop copy
- USB-C debug (CH340C), UART header and test points
- 4-layer board, 100 × 100 mm maximum

Applications

- Power, CAN and E-Stop distribution on Sooner Competitive Robotics' IGVC 2027 robot
- Per-board current and voltage monitoring for the team

Online version: https://saosith.github.io/projects/scr-hub-2027/

Description

The Hub sits between the robot's battery and every other board. It distributes 12 V power, the CAN bus and the hardware E-Stop line to seven 6-pin Molex ports, and it measures the current and voltage on each port so the team can see what every board draws. I redesigned it for 2027 around a proposed STM32C092 microcontroller, P-MOSFET reverse-polarity protection, a 3.3 V buck converter and seven INA220 current sensors on a 4-layer board. Status: schematic complete; PCB layout in progress.

Team project with Sooner Competitive Robotics (IGVC 2027). My role: schematic, part selection and PCB layout of the Hub.

Key parameters

Parameter	Value
Input	12 V, 5-8 A total
Board ports	7 with current sensing
Board	4-layer 100 x 100 mm max
Status	layout in progress

Data status: schematic complete, PCB layout in progress. No values on this page are measured.

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1 Introduction

The Hub is the board every other board on our robot plugs into. It takes power from the battery and passes three things to each board: 12 V power, the CAN bus and the E-Stop line. For 2027, the team wanted more from it: current and voltage sensing for every board, a temperature reading for fan control, and a measure of how busy the CAN bus is [1]. I started from last year's Hub, reviewed what to keep, and redesigned the power input, the 3.3 V supply, the sensing and the debug access.

Status: schematic complete. **Next:** finish placement and routing, then a team and captain review before ordering. No values on this page are measured.

2 Requirements

- **Distribution:** power, CAN and the E-Stop line to every board through 6-pin Molex connectors [1].
- **Sensing:** current and voltage for every board, plus CAN bus utilization [1]. Temperature sensing for the fan is shared with the Fan Controller board; ownership is still open.
- **Load:** about 5-8 A total for the boards the Hub feeds (team estimate), not counting the motors.
- **Safety:** the E-Stop path must be hardware only, never through firmware [2].
- **Input protection order:** fuse, then a TVS diode in parallel, then series reverse protection [2].
- **Board:** 4-layer boards must stay within 100 x 100 mm [2]. Parts should be surface-mount, hand-solderable, and stocked at DigiKey or JLCPCB/LCSC.

3 System overview

Figure 1 shows the board in three layers. The **power layer** runs from the XT60 through the input protection to a 12 V bus that feeds the seven board ports and the 3.3 V buck. The **data layer** is the CAN bus to every port, plus an I²C bus from the microcontroller to the seven current sensors. The **safety layer** is the E-Stop line, which a buffer copies to every port with no firmware in the path.

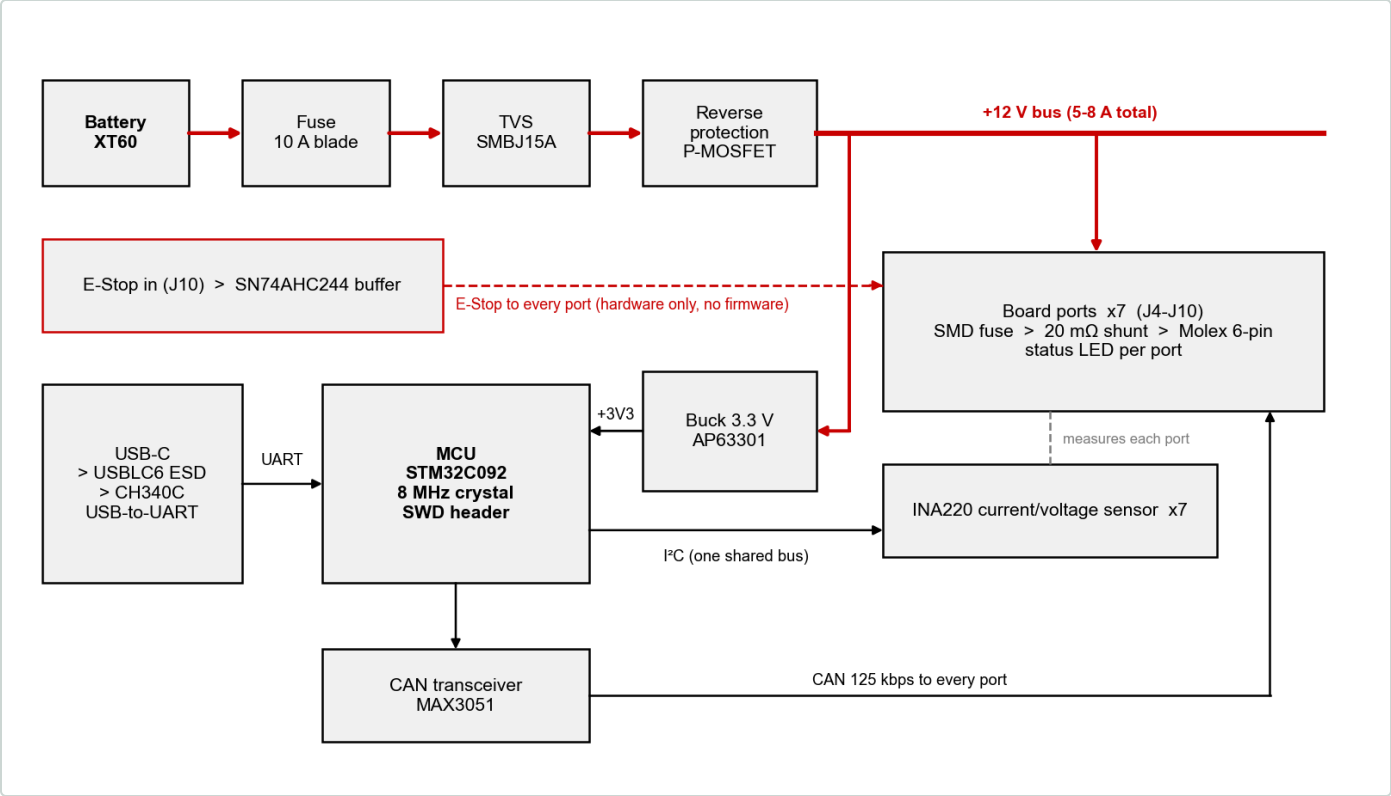


Fig. 1 Hub block diagram. Red lines carry 12 V power; the dashed red line is the hardware E-Stop path.

4 Design

4.1 Input protection

Battery power enters through an XT60 and passes a 10 A blade fuse in a Keystone 3557-2 holder, then an SMBJ15A TVS diode that clamps voltage spikes to ground. Normally the team uses a series Schottky diode for reverse-polarity protection [2]. At 8 A, a Schottky drops about 0.45 V and dissipates about 3.6 W. I used a P-MOSFET instead: the DMP4015SK3Q-13 has 11 mΩ of on-resistance at V_{GS} = −10 V [3], so $P = I^2 \times R = 8^2 \times 0.011 \approx 0.7$ W.

The drain faces the fuse and the source faces the 12 V bus, so the body diode conducts first and the channel then turns on. If the battery is connected backwards, the gate cannot turn on and the MOSFET blocks. A 10 kΩ resistor pulls the gate to ground, and a 12 V Zener clamps the gate-source voltage within the ±25 V rating. A 100 µF bulk capacitor, a 10 µF ceramic and a 0.1 µF ceramic sit on the bus.

4.2 3.3 V supply

Last year's Hub used an LDO, which turns the whole 12 V to 3.3 V drop into heat. I replaced it with an AP63301 synchronous buck converter (3.8-32 V input, 3 A, 500 kHz) [4]. The component values follow the datasheet's 3.3 V table: R12 = 93.1 kΩ and R13 = 30.1 kΩ for feedback, 56 pF feed-forward, a 4.7 µH inductor, a 0.1 µF bootstrap capacitor, 10 µF at the input and 2 × 22 µF at the output. Because the converter is synchronous, no catch diode is needed.

4.3 Board ports and current sensing

Each of the seven ports has the same chain: an SMD fuse, a 20 mΩ shunt resistor, then pin 1 of the Molex connector. Pin 6 of each connector drives a status LED through 1 kΩ. An INA220 measures the voltage across the shunt and the port voltage [6]. At 2.5 A the shunt drops 50 mV, inside the INA220's ±80 mV range, and its 10 µV step gives 10 µV / 20 mΩ = 0.5 mA resolution. The sense lines are routed as Kelvin connections from the shunt pads. All seven sensors share one I²C bus, with unique addresses set by strapping A0 and A1.

4.4 Microcontroller, clock and CAN

The STM32C092 has a built-in CAN controller (FDCAN), so it needs only a MAX3051 transceiver. This matches last year's Hub and Safety Lights boards, so their firmware can be reused. The internal HSI48 oscillator can drift ±1% over 0-85 °C [5], and two nodes drifting in opposite directions can break CAN timing, so the CAN clock comes from an 8 MHz crystal (Abracon ABLS-8.000MHZ-B4-T). At 125 kbps that gives 8 MHz / 125 kbps = 64 clock cycles per bit. The crystal's load capacitance is C_L = 18 pF, so the load capacitors are C = 2 × (C_L − C_{stray}) = 2 × (18 − 10) = 16 pF [5]. Bus termination is a 120 Ω resistor with a jumper, so the Hub can be either end of the bus.

4.5 E-Stop path

The E-Stop line is active-low 3.3 V logic: high means the robot may run. It comes in from the E-Stop relay board on J10, so J10's E-Stop pin is the input, and an SN74AHC244 buffer drives a separate copy to the other six ports (J4–J9). Firmware never touches this path.

4.6 Debug access

A USB-C connector feeds a USBLC6-2SC6 ESD array and a CH340C USB-to-UART bridge, which connects to the microcontroller's UART through 1 kΩ resistors. A 3-pin UART header sits on the microcontroller side of those resistors, so an external adapter can still be used. Test points cover +12 V, +3.3 V, CANH, CANL, the E-Stop line, SDA, SCL and ground.

5 Design decisions

Decision	Options considered	Choice	Why	Status
Reverse protection	Series Schottky (team guide default), or a P-MOSFET	**P-MOSFET DMP4015SK3Q-13**	At 8 A a Schottky drops about 0.45 V and dissipates about 3.6 W. The MOSFET dissipates about 0.7 W. I chose it for efficiency and less heat.	decided
3.3 V supply	Last year's LDO, or a buck converter	**AP63301 buck**	An LDO's efficiency is at most V_out / V_in = 3.3 / 12 ≈ 28% here. At the Hub's estimated 50–100 mA load, the AP63301 datasheet curve for 12 V to 3.3 V shows about 55–70%, and it accepts up to 32 V.	decided
Microcontroller and CAN	Pico + MCP2515 (team standard), or STM32C092 + MAX3051	**STM32C092 + MAX3051**	One chip handles CAN, it matches the 2026 Hub and Safety Lights, and their firmware can be reused.	open
MCU clock	Internal HSI48 oscillator, or an 8 MHz crystal	**8 MHz crystal, Abracon ABLS (16 pF load caps)**	HSI48 can drift ±1% over temperature, and two nodes drifting apart can break CAN timing. The ABLS is ±30 ppm and has a footprint in the library.	decided
I²C buses	One shared bus, or two buses for fault isolation	**One bus**	Standard for about 8 devices; firmware timeouts and bus recovery handle a stuck device.	open
Debug access	None, test points only, UART header, or on-board USB-C	**USB-C (CH340C) + UART header + test points**	One cable for debugging, with the header as a backup and for probing.	decided
Layer count	2-layer, or 4-layer	**4-layer**	A solid ground plane makes the buck, USB and CAN layout forgiving for a first board and frees routing space.	decided

Main fuse rating	10 A, or 15 A	**10 A for now**	Depends on whether the NUC computer is powered through the Hub (about 2 A more).	open

6 Layout

The board is 4-layer: signals on L1 and L4, a solid ground plane on L2, and split +12 V / +3.3 V regions on L3, on PCBWay's 1.6 mm stack-up [7]. The design rules keep a margin over PCBWay's minimums: 0.2 mm clearance, 0.3 mm / 0.6 mm vias, 0.5 mm copper to the board edge, and wider minimum widths for the power nets (1.0 mm for port power, 1.5 mm for +12 V). USB D+/D− is routed as a 90 Ω differential pair.

The Molex ports sit around the edges with their port blocks (fuse, shunt, INA220) right behind them. The XT60 input, fuse holder and P-MOSFET are on the right, the microcontroller is in the center with the crystal at its pins, the USB-C connector is on the bottom edge, and the buck converter is in the bottom-right corner, away from the crystal and the USB lines. The 10 A input path is drawn as wide pours on the outer layers.

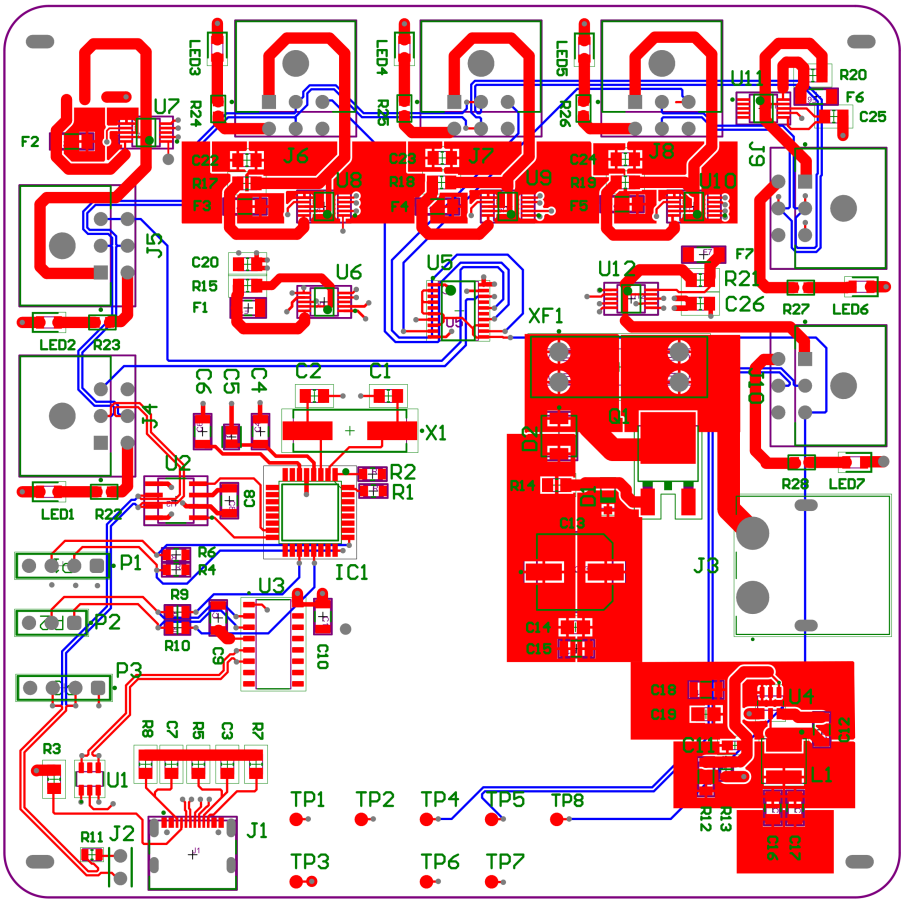


Fig. 2 Altium PCB print, work in progress: top copper in red, bottom copper in blue. Routing is not finished.

7 Project status

Sections for simulation and test results are added as each stage is completed.

Stage	Status	Plan

Requirements and architecture	✓ Complete	From the 2027 task document and a review of last year's Hub.
Schematic	✓ Complete	Five sheets, in the Altium PDF. Small fixes listed under Open issues.
Design rules and stack-up	✓ Complete	PCBWay 4-layer 1.6 mm stack-up; rules with margin over PCBWay minimums.
Placement and routing	▶ In progress	Power path as pours, buck loop, Kelvin sense traces, USB and CAN pairs.
Design review	○ Planned	Team and captain review before any order.
Fabrication	○ Planned	PCBWay, 4-layer 1.6 mm, after captain approval.
Firmware	○ Planned	Start from the 2026 Safety Lights code (same STM32C092); report port currents over CAN.
Bring-up and test	○ Planned	Rail voltages, reverse-polarity test, current readings against a bench meter, CAN and E-Stop checks.

8 Open issues

- **Main fuse rating.** 10 A, or 15 A if the NUC is powered through the Hub. Waiting on the team's power plan.
- **Port fuse ratings.** Each board's maximum current is still needed to size its fuse. Port 1 uses a Littelfuse 0466005.NR while ports 2-7 use SF-1206S100-2; these should be reconciled.
- **Spare E-Stop buffer inputs.** Pins 15 and 17 of the SN74AHC244 are marked no-connect; tie them to ground.
- **Port surge protection.** The ports have fuses but no TVS diodes; decide whether each port needs one.
- **Overlap with the Sensor Board.** Both boards measure the main current and temperature; the team needs to decide which board owns what.
- **Approvals.** The STM32C092, the single I²C bus, the P-MOSFET and the PCBWay order all need captain approval.

9 Revision history

Rev	Date	What changed
A	2026-09-24	Project started; requirements collected from the team documents.
A	2026-10-01	Schematic started from last year's Hub; crystal and buck converter chosen.
A	2026-10-04	Reverse protection changed from a Schottky to a P-MOSFET.
A	2026-10-05	Buck changed to the AP63301; on-board USB-C to UART added.
A	2026-10-08	4-layer board chosen; PCBWay stack-up and design rules set (10-09 to 10-10).
A	2026-10-12	Placement and routing in progress; main input path drawn as pours.

10 References

1. Sooner Competitive Robotics, IGVC 2027 Electrical Requirements and Tasks
2. Sooner Competitive Robotics, Board Design Guide, Rev 0.5
3. Diodes Inc., DMP4015SK3 P-channel MOSFET datasheet
4. Diodes Inc., AP63300/AP63301 synchronous buck converter datasheet
5. STMicroelectronics, STM32C092 datasheet, and AN2867 oscillator design guide

6. Texas Instruments, INA220 current and power monitor datasheet
7. PCBWay, 4-layer 1.6 mm standard stack-up

Appendix: Bill of materials

Ref	Part	Value	Package	Source
J3	Amass XT60PW-F battery connector	—	XT60 right-angle	TBD
XF1	Keystone 3557-2 fuse holder (10 A blade fuse)	—	ATO clip	LCSC C352820
D2	SMBJ15A input TVS	—	SMB	TBD
Q1	Diodes Inc. DMP4015SK3Q-13 P-MOSFET	—	TO-252	LCSC C461089
D1	Diodes Inc. BZT52C12-7-F gate Zener	12 V	SOD-123	LCSC C124196
U4	Diodes Inc. AP63301WU-7 buck converter	3.3 V out	TSOT-26	TBD
IC1	ST STM32C092KBT6 microcontroller	—	LQFP-32	TBD
X1	Abracon ABLS-8.000MHZ-B4-T crystal	8 MHz	HC-49/US SMD	LCSC C596838
U2	Maxim MAX3051ESA+T CAN transceiver	—	SOIC-8	TBD
F1-F7	Port fuses: Littelfuse 0466005.NR (F1), SF-1206S100-2 (F2-F7)	—	1206	TBD
R15-R21	Yageo RL1206FR-070R02L port shunt	20 mΩ	1206	LCSC C513558
U6-U12	TI INA220AIDGSR current sensor	—	VSSOP-10	TBD
U5	TI SN74AHC244QPWRQ1 E-Stop buffer	—	TSSOP-20	TBD
LED1-LED7	AP2012SGD3 port status LED	—	0805	TBD
J1	GCT USB4105-GF-A USB-C receptacle	—	SMD	LCSC C3020560
U1	ST USBLC6-2SC6 USB ESD array	—	SOT-23-6	LCSC C7519
U3	WCH CH340C USB-to-UART bridge	—	SOIC-16	LCSC C84681
J4-J10	6-pin Molex board connectors	—	—	TBD